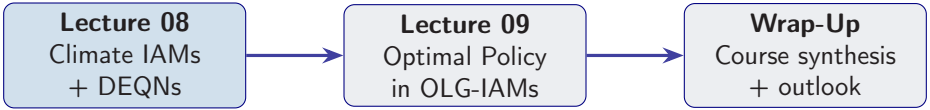
Lecture 27 (B26): Climate Economics and Integrated Assessment Models

Why IAMs, social cost of carbon, DICE / CDICE baseline

Simon Scheidegger

HEC, University of Lausanne

Climate Block Roadmap



This lecture (∼75 min):

1. Motivation: why climate economics? (∼12 min)
2. The DICE/CDICE climate model (∼18 min)
3. A stochastic IAM: full model and derivation (∼25 min)
4. Deep uncertainty quantification (∼18 min)

Part I

Why Climate Economics?

Climate Change as an Economic Threat

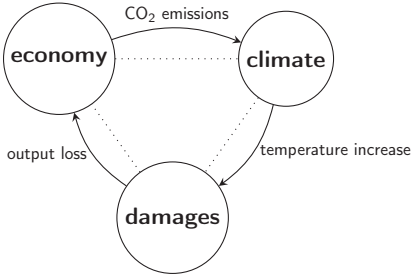
Observed global warming motivates the IAM feedback loop below.

- Global warming is a growing threat to economic well-being.
- Tipping points place us in catastrophic danger.
- Impacts current and future generations in different regions very differently.
- Recent overviews: Dietz (2024), van der Ploeg and Rezai (2026), Fernandez-Villaverde, Gillingham, Scheidegger (2025).

Why Integrated Assessment Models (IAMs)?

IAM logic

Couple the economy (output, investment, abatement, welfare) with the climate system (carbon concentrations, temperature, damages).



Output of interest: social cost of carbon (SCC), optimal tax paths, welfare decomposition.

The Social Cost of Carbon (SCC)

Definition

SCC at time t: marginal welfare cost of one extra unit of emissions, converted to consumption units.

$$\mathrm{SCC}_t = -\frac{\partial V_t / \partial E_t}{\partial V_t / \partial C_t}$$

High SCC when

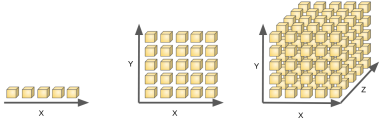
- Damages are steep
- Climate response is strong
- Discounting is low
- Tipping risk is material

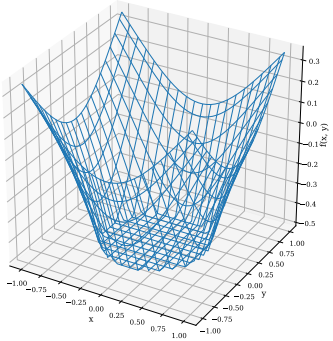
Policy interpretation

- Tax path for carbon pricing
- Benchmark for regulation / cap-and-trade
- Input to cost-benefit analysis
- Why uncertainty matters: SCC is a distribution

Why This Is Computationally Hard

- **Nonstationarity**: TFP, population, emissions intensity, and forcing paths all change exogenously over time.
- **Coupled dynamics**: economy and climate interact in both directions.
- **Long horizon**: welfare effects unfold over 100–300 years.
- **Curse of dimensionality**: nonlinearities, tipping hazards, constraints.





Our Contribution

- **Generic global solution method** to efficiently compute global solutions to stochastic IAMs that include economic and climate risks.
- **Perform global sensitivity analysis** with respect to uncertain parameters most discussed in the literature.

1. Exploit **deep equilibrium nets** (DEQNs) to alleviate the curse of dimensionality.
2. Add model parameters as **pseudo-states** $\rightarrow$ solve the model only once.
3. Construct **surrogate models** for the SCC (and other QoIs) using **Gaussian Process regression** and **Bayesian active learning**.

Part II

The DICE/CDICE Climate Model

DICE Baseline: Minimal Global IAM

State variables

- Economic stock: K_t
- Carbon stocks: $M_t^{\mathrm{AT}}, M_t^{\mathrm{UO}}, M_t^{\mathrm{LO}}$
- Temperature states: $T_t^{\mathrm{AT}}, T_t^{\mathrm{OC}}$
- Exogenous trends: population L_t, TFP A_t, emissions intensity σ_t

Controls

- Savings / investment share s_t
- Abatement rate μ_t

Why DICE is influential

- Transparent
- Fast to evaluate
- Policy-facing
- Benchmark for extensions

Challenge

Fast is useful, but only if the climate module is calibrated consistently with climate science.

Carbon Cycle (3-Box Model)

- DICE-2016 models the carbon cycle via three carbon reservoirs:
  - Atmosphere (AT), Upper ocean (UO), Lower ocean (LO)
- Aggregate distribution of carbon: $M_t = (M_{\mathrm{AT},t}, M_{\mathrm{UO},t}, M_{\mathrm{LO},t})^T$

Carbon concentrations evolve as a linear dynamic system:

$$M_{t+1} = (I + \Delta_t \cdot B) \cdot M_t + \Delta_t \cdot E_t$$

- Carbon emissions (economic activity + exogenous source): $E_t = E_{\mathrm{ind},t} + E_{\mathrm{Land}}$
- Industrial emissions: $E_{\mathrm{ind},t} = (1 - \mu_t)\sigma_t K_t^{\alpha}(A_t L_t)^{1-\alpha}$
- B is the transfer matrix governing exchange rates between reservoirs

PAUSE Finger Exercise: One-Step Carbon Cycle

Exercise (5 min)

The simplified carbon cycle has atmospheric CO_2 evolving as:

$$M_{\mathrm{AT}}^{t+1} = E_t + \phi_{11}\, M_{\mathrm{AT}}^{t} + \phi_{21}\, M_{\mathrm{UP}}^{t},$$

where $\phi_{11} = 0.912$, $\phi_{21} = 0.0444$, and M_{UP}^{t} (upper ocean) is given.

1. With $M_{\mathrm{AT}}^{2020} = 851$ GtC, $M_{\mathrm{UP}}^{2020} = 628$ GtC, and emissions $E_{2020} = 10$ GtC/yr (over a 5-year step: $E = 50$ GtC): compute M_{AT}^{2025}.
2. If emissions doubled to $E = 100$ GtC, what would M_{AT}^{2025} be?
3. By how many GtC does doubling emissions increase atmospheric CO_2?

Solution: One-Step Carbon Cycle

Answer

1. $M_{\text{AT}}^{2025} = 50 + 0.912 \times 851 + 0.0444 \times 628 = 50 + 776.1 + 27.9 = 854.0$ GtC.
2. $M_{\text{AT}}^{2025} = 100 + 776.1 + 27.9 = 904.0$ GtC.
3. Doubling emissions adds $904.0 - 854.0 = 50$ GtC—exactly the extra emissions. The carbon cycle is linear in emissions (at one-step horizon), so the effect is additive.

Temperature Dynamics (2-Box Model)

The two-layer energy balance model in DICE-2016:

$$T_{t+1}^{\mathrm{AT}} = T_t^{\mathrm{AT}} + \Delta_t \cdot c_1 \left(F_t - \lambda T_t^{\mathrm{AT}} - c_3 \left(T_t^{\mathrm{AT}} - T_t^{\mathrm{OC}}\right)\right)$$
$$T_{t+1}^{\mathrm{OC}} = T_t^{\mathrm{OC}} + \Delta_t \cdot c_4 \left(T_t^{\mathrm{AT}} - T_t^{\mathrm{OC}}\right)$$

Radiative forcing:

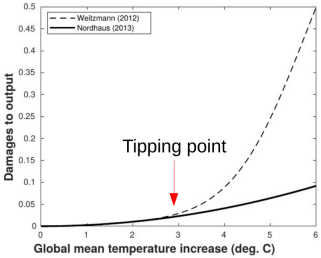
$$F_t = F_{2\times\mathrm{CO}_2} \frac{\log(M_t^{\mathrm{AT}}/M_{\mathrm{EQ}}^{\mathrm{AT}})}{\log(2)} + F_t^{\mathrm{EX}}$$

Key parameters

$\lambda = F_{2\times\mathrm{CO}_2}/\Delta T_{\mathrm{AT},\times 2}$ where $\Delta T_{\mathrm{AT},\times 2}$ is the equilibrium climate sensitivity (ECS) — the long-run warming from CO_2 doubling.

Damage Functions

See also Tol (2009), Hänsel et al. (2020)



Retained-output damage factors decline as warming rises.

Nordhaus (2013):

$$\Omega_t^N(T_{\mathrm{AT},t}) = \frac{1}{1 + \pi_1 T_{\mathrm{AT},t} + \pi_2 T_{\mathrm{AT},t}^2}$$

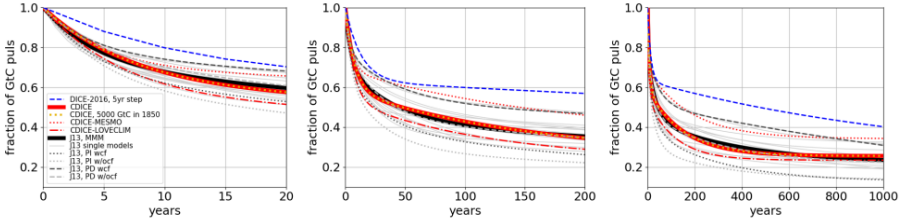
Weitzman (2012) — with tipping:

$$\Omega_t(T_{\mathrm{AT},t}) = \frac{1}{1 + \left(\frac{1}{\psi_1} T_{\mathrm{AT},t}\right)^2 + \left(\frac{1}{2 \cdot TP_t} T_{\mathrm{AT},t}\right)^{6.754}}$$

- Degree of convexity is of key importance in determining the optimal taxes [Weitzman 2012].
- Tipping points: losing the Amazon rain forest, faster onset of El Niño, reversal of the Gulf Stream, etc.

Why Recalibrate? → CDICE

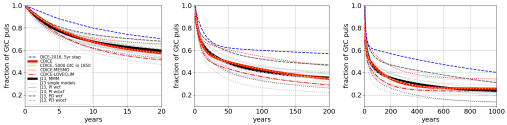
- Climate emulators must reproduce benchmark behavior from climate science model archives (CMIP).
- Earlier critiques (Dietz et al., 2021) highlighted major discrepancies for common IAM calibrations.
- Folini et al. show that the functional form can remain simple, but parameters require systematic calibration.



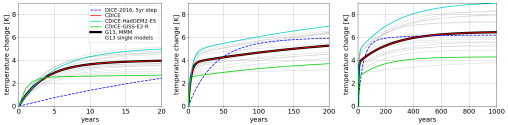
CDICE recalibrates the reduced-form climate emulator to climate-science benchmarks.

Four-Test Framework (Folini et al., 2024)

Test	Targets	Use
1. Carbon pulse (100 GtC)	Atmospheric retention path	Calibrate carbon cycle
2. $4\times CO_2$ step	Temperature impulse response	Calibrate temperature block
3. 1% CO_2/year	Transient climate response (TCR)	Out-of-sample validation
4. Historical + RCP	Realistic forcing paths	End-to-end validation



Carbon-pulse response.



Temperature response to a $4\times CO_2$ forcing step.

Equilibrium Climate Sensitivity (ECS) — A Key Uncertainty

IPCC AR5 gives a 'likely range' of [1.5K, 4.5K]

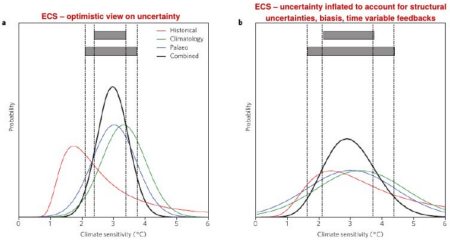


Figure 5 | Illustrative example of combining multiple constraints for climate sensitivity. Overall PDFs are the products of three PDFs based on the historical warming, climatological constraints on mostly GCMs, and palaeoclimate. Grey ranges at the top indicate a 'likely' (66%) and 'very likely' (90%) combined range. **a,b**. Constraint based on an optimistic interpretation of uncertainty ranges and assuming full independence (**a**) and on inflated ranges (**b**) to account for structural uncertainties, and with historical estimates inflated and scaled up to account for observation biases, and feedbacks varying from historical to future and across forcings. See Methods for details.

Knutti et al. 2017, Nature, doi: 10.1038/NGEO3017

- Roe and Baker (2007): $\mathrm{ECS} = \frac{\lambda_0}{1-f} F_{2\times \mathrm{CO}_2}$, $f \sim \mathcal{N}(\mu_f, S_f)$.
- Making ECS stochastic is a natural channel for Bayesian learning.

Transparent Time-Step Formulation

A practical contribution in CDICE: expose the time-step scaling directly in equations.

$$X_{t+\Delta t} = X_t + \Delta t \cdot f(X_t, u_t; \theta)$$

- Allows coherent annual, 5-year, or 10-year implementations within one generic formulation.
- Time discretization consistency improves transferability between classroom code, production code, and policy simulation horizons.

Summary of climate module:

Climate states (5)	$M_{\mathrm{AT}}, M_{\mathrm{UO}}, M_{\mathrm{LO}}, T^{\mathrm{AT}}, T^{\mathrm{OC}}$
Key parameters	ECS ($\Delta T_{\mathrm{AT},\times 2}$), carbon transfer B, forcing $F_{2\times \mathrm{CO}_2}$
Uncertainty channels	ECS, damage curvature, tipping